

TrES-3b

R-2168

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_121019 · created 2026-08-13T00:20:21+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456219.6415 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.057
Transit depth (Rp/Rs)^2	3.81 +/- 2.21 [%]
Orbital Inclination (inc)	86.53 +/- 2.88
Airmass coefficient 1 (a1)	0.65 +/- 0.13
Airmass coefficient 2 (a2)	0.28 +/- 0.15
Scatter in the residuals of the lightcurve fit is	20.72 %
Best Comparison Star	#1 - [586, 356]
Optimal Aperture	1.36
Optimal Annulus	9.28
Transit Duration (day)	0.078 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.057 vs 0.16309 (deviation 0.56σ)
Duration fit vs published	0.078 d vs 0.0567 d (deviation 1.64σ)
Mid-time O–C	-3.1 min
Depth SNR	3.4
Residual scatter	20.72 %

Light curve

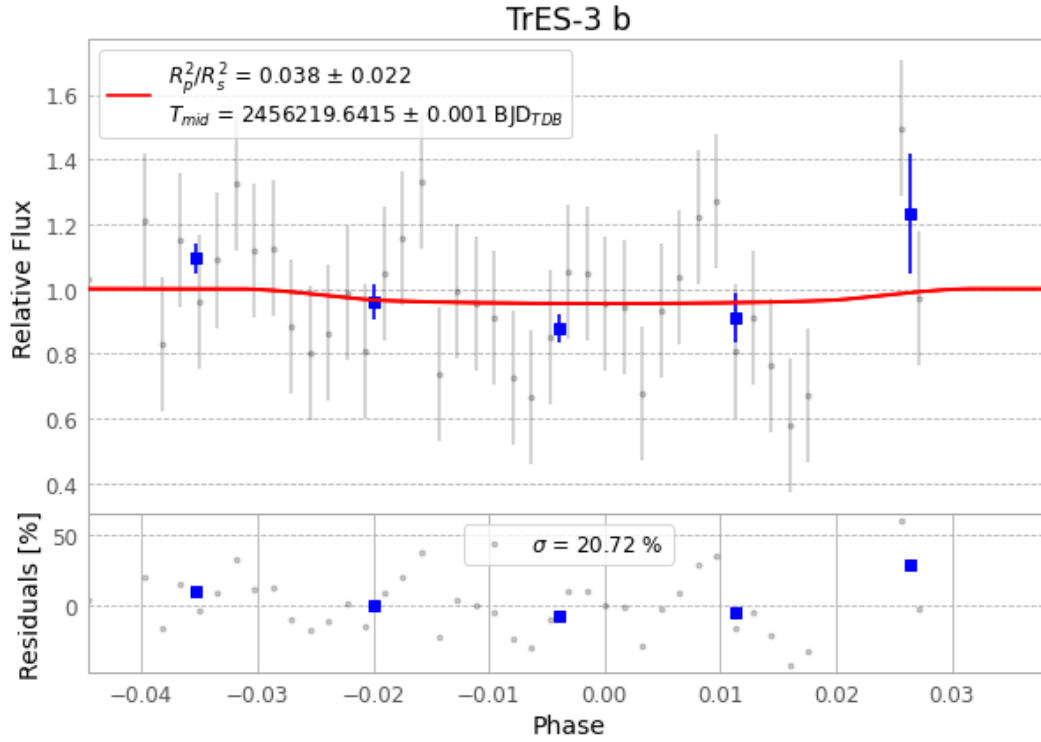


Figure 1 — FinalLightCurve_TrES-3 b_2012-10-18.png (SHA-256 cabcd4070b8635aa1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [265, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9bdc098e97693136...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

60 FITS frames (39 MB), TRES-3121019020014.FITS → TRES-3121019050312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-121019.log (9715bbb95b1a...), FinalLightCurve_TrES-3 b_2012-10-18.png (cabcd4070b863...), FinalLightCurve_TrES-3 b_2012-10-18.csv (f1bef898232d...)

Compute resources

Wall time 194.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:20Z.